Data Communications Planning

Quick briefing

Define who must exchange what information, how fast and how securely; match fixed, tactical and special purpose technologies to terrain and threat; apply the signal planning process with PACE and recovery; and size the data networks that carry voice.

Know the facts

C4S · Command and Control Communications, and Cyber Systems: the capabilities that enable signal. Its four sub-systems are communications, electronics, information and cyber (PAM 6-00 0127, 0401).

Connect, Sustain, Recover · Connect gives connectivity to all Army units, Sustain keeps it going through the operation, and Recover repairs and restores it when attacked or destroyed (0144).

Data transmission requirement · A statement of who must exchange what information, with whom, how quickly, how securely and over what terrain. It comes from the mission and the commander, not from the equipment on hand.

Requirement categories · Reliability and redundancy, security, speed and capacity, interoperability, scalability, coverage and range, mobility, and environmental resilience and sustainment.

PANET and the AFP FCS · PANET is the secured, managed Army intranet, with its primary hub at the PA communications center, Fort Bonifacio. The AFP FCS is the transmission superhighway between switches, run by GHQ-AFP.

Recovery teams · The TOC team, the C3 restoration team, the network recovery team and the cyber security incident response team (CSIRT) execute recovery (0143, 0228 to 0231).

Voice sizing figures · About 80 to 90 kbps per uncompressed G.711 call and about 25 to 30 kbps per compressed G.729 call, plus the data, plus a margin of at least 25 percent.

Understand the reasoning

Requirements come first and equipment second. Start from the information exchanges the mission needs, then see what the resources can support and rank the rest.

Once voice rides on a data network, the voice plan is only as reliable as that network, including its power, antennas, switches and operations center.

No single technology works everywhere, so each link is matched to its terrain, spectrum and threat, and its fallback must not share the same failure point.

Philippine Army classroom exercise

In a fictional Philippine Army classroom exercise, a task force TOC reaches higher headquarters through one microwave link. A battalion operates on an island across open water, and the higher headquarters now wants a daily video-teleconference and drone video from the TOC.

Decision · What should the data plan change, and how should the TOC-to-higher link be checked?

Worked analysis · One microwave link fails the redundant and survivability principles, so the plan adds an independent path, such as a satellite terminal, and keeps HF/SSB as the secondary means of the PAM 6-00 0406 table. The island battalion needs HF, satellite or relay stations because line of sight fails over water. The new video and VTC traffic is stated as requirements with volume, delay, precedence and classification, then the link is sized: voice calls times the per-call figure, plus data, plus at least 25 percent margin. If the link is short, reduce the load at peak hours or ask for more capacity.

How to practice

1 · List each information exchange requirement with originator, recipient, information type, volume and frequency, maximum delay, precedence, classification and phase.

2 · Check that each requirement is specific, measurable, tied to the mission or CCIR, ranked and phased.

3 · Choose the means for each link from the 0406 table and the environment responses, and note each means' limits.

4 · Write the PACE table for each critical service, including voice, so that no two levels share a failure point, and add recovery procedures for each contingency.

5 · Size the links that carry voice and data, finalize against the checklist, and attach the data communications appendix to the signal plan.

Check yourself

Name four sources of data transmission requirements for the signal planner. · Any four of: commander's guidance or intent, staff guidance, higher, subordinate and adjacent units, the SOP, and the CEOI and CESI.

What must the plan include for compromised frequencies after captured equipment? · Procedures to change frequencies efficiently, in the shortest possible time (0311).

Why is a voice service that runs over IP not made safe just by having a backup router? · Power, antennas, switches and the network operations center must also survive, and the fallback voice path must not share the same failure point.

Course source

Data Communications Planning · Student Handout